# Decoding music preference from EEG

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**Abstract**

This project investigates whether binary music preference (like versus dislike) can be decoded from electroencephalography (EEG) recorded while participants listened to one-minute music video excerpts. Using the DEAP dataset (32 subjects, 40 trials each), we extract a 164-dimensional feature set combining differential entropy across five frequency bands with frontal alpha asymmetry (FAA), and evaluate three classification strategies under 5-fold stratified cross-validation. The best strategy achieves a mean within-subject accuracy of 64.5% using margin-based label binarization. SHAP analysis reveals that all four FAA features occupy the top four positions globally, providing strong support for the approach-withdrawal hypothesis. Time-resolved decoding shows preference information is first decodable within 2.5s and peaks at 61.0% near 25.0 seconds. Both high- and low-familiarity conditions exceed the overall baseline, with high-familiarity trials reaching 71.2%.

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# 1 Introduction & background

Music preference is a subjective phenomenon with measurable correlates in neural activity. When a person hears music they enjoy, changes in cortical activity appear within seconds, particularly in frontal regions associated with reward and approach motivation. Decoding these preferences directly from EEG signals would have applications in passive music recommendation and the neuroscience of aesthetic experience.

The theoretical backbone of this work is the approach-withdrawal model of Davidson (1992), which claims that preferred stimuli are associated with greater left frontal cortical activation, while disliked stimuli engage the right hemisphere. In EEG, this asymmetry is visible in the alpha band (8–13 Hz) through the frontal alpha asymmetry (FAA) index:

$$\mathrm{FAA} = \ln(\alpha_{\mathrm{right}}) - \ln(\alpha_{\mathrm{left}}).$$

A positive FAA value indicates greater left frontal activation. We hypothesize that FAA features will rank among the most predictive of the feature set.

Our pipeline consists of four stages: (1) data exploration and binarization of the continuous liking scale, (2) feature engineering, (3) within-subject cross-validated classification, and (4) model interpretability.

# 2 Data & features

## 2.1 Dataset

The DEAP dataset (Koelstra et al. 2012) contains physiological recordings from 32 participants (16 male, 16 female; ages 19–37) while they watched 40 one-minute music video excerpts. After each excerpt, participants rated their emotional response on four 9-point continuous scales: valence (unhappy to happy), arousal (calm to excited), dominance (submissive to in-control), and liking (dislike to like). Only the liking dimension is used as the target variable.[1]

Table 1 summarizes descriptive statistics for all four dimensions across all 1,280 trials. We binarize continuous liking scores via a median split per-subject, yielding a near-balanced class distribution (638 like, 642 dislike). A second strategy, margin binarization, removes trials within ±0.5 rating units of the median to exclude ambiguous labels at the cost of reduced sample size (e.g., subject 1 retains 27 of 40 trials, removing 32%).

[1] EEG was recorded with a 32-channel BioSemi ActiveTwo cap following the 10–20 system at 512 Hz, then downsampled to 128 Hz. Each trial consists of a 3-second baseline epoch followed by a 60-second stimulus epoch, yielding 7,680 samples per trial and 1,280 trials in total.

Table 1: Descriptive statistics of self-report ratings across all subjects and trials ($n = 1{,}280$).

| Dimension | Mean | Std | Min | Max | Median |
|---|---|---|---|---|---|
| Valence | 5.25 | 2.13 | 1.0 | 9.0 | 5.04 |
| Arousal | 5.16 | 2.02 | 1.0 | 9.0 | 5.23 |
| Dominance | 5.38 | 2.10 | 1.0 | 9.0 | 5.24 |
| Liking | 5.52 | 2.28 | 1.0 | 9.0 | 6.05 |

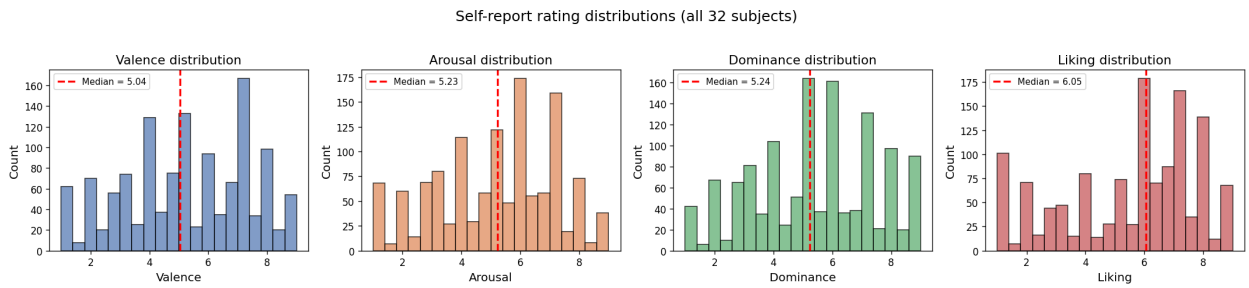


Figure 1: Distributions of self-report ratings across all 1,280 trials. All four dimensions span the full 1–9 range.

The 32 electrodes span frontal, central, parietal, occipital, and temporal regions (Figure 2). Frontal coverage is particularly important for FAA, which relies on paired symmetric electrodes. Figure 3 shows a representative power spectrum.

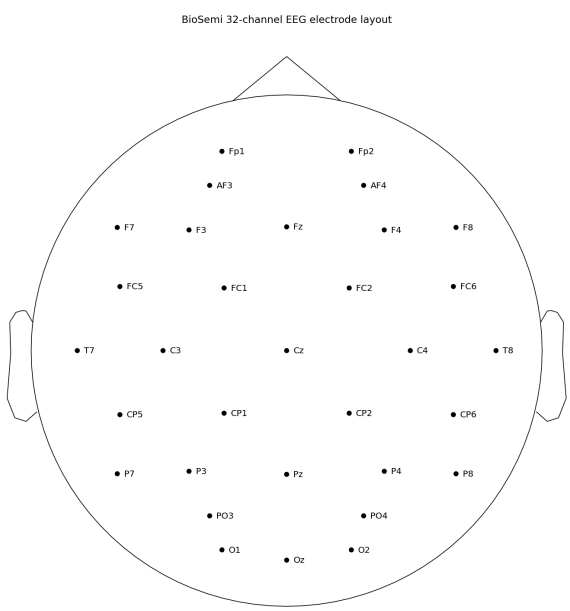


Figure 2: BioSemi 32-channel electrode layout. Frontal pairs used for FAA are Fp2/Fp1, AF4/AF3, F4/F3, and F8/F7.

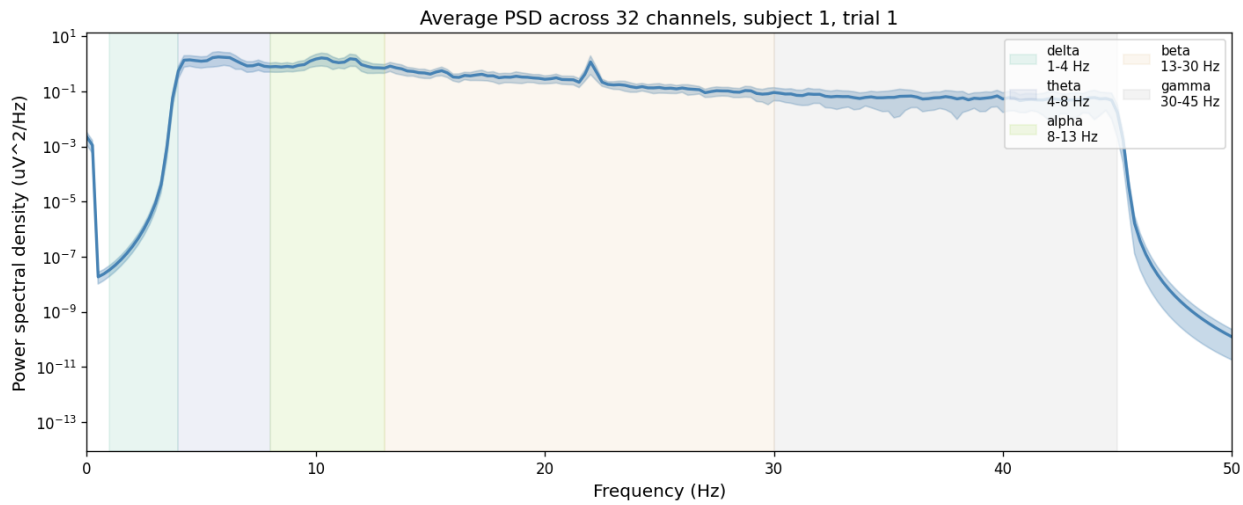


Figure 3: Mean power spectral density averaged across all channels for subject 1, trial 1.

## 2.2 Feature engineering

Differential entropy (DE) quantifies the information content of a continuous signal. Under the assumption of a Gaussian distribution, it reduces to:

$$\mathrm{DE}(x) = \tfrac{1}{2}\ln(2\pi e\,\sigma^2),$$

where $\sigma^2$ is the band-limited power estimated via Welch's method. DE is computed for each of the 32 channels across five frequency bands (delta 1–4 Hz, theta 4–8 Hz, alpha 8–13 Hz, beta 13–30 Hz, gamma 30–45 Hz), yielding a 160-dimensional feature vector. DE has been shown to outperform raw band power for EEG-based emotion recognition (Duan, Zhu and Lu 2013). Figure 4 shows a representative DE heatmap.

The FAA index is computed for four symmetric frontal electrode pairs (Fp2/Fp1, AF4/AF3, F4/F3, F8/F7), each contributing one feature. Figure 5 shows the distribution of FAA values by preference label. Combined with DE, the full feature vector has 164 dimensions.

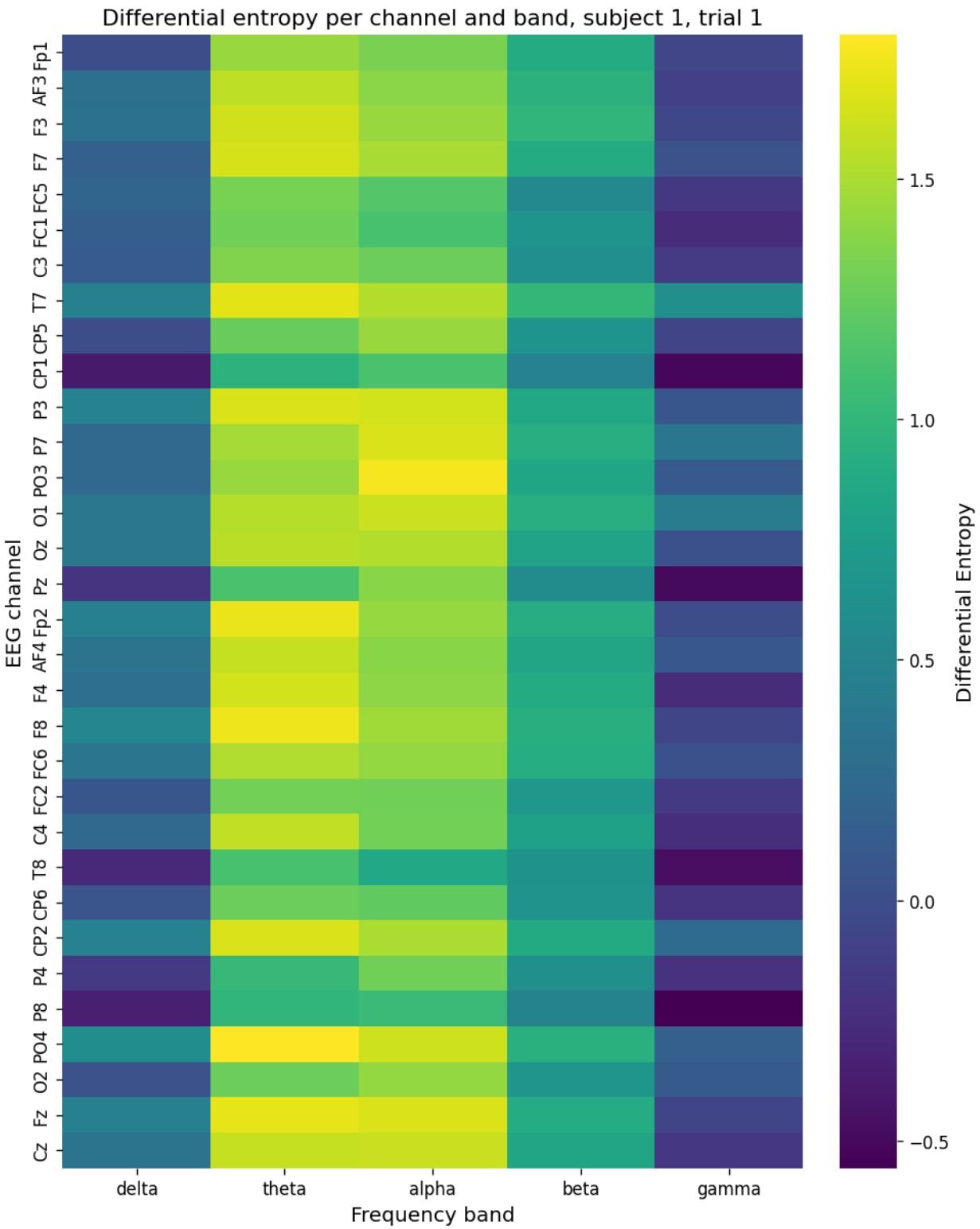


Figure 4: Differential entropy heatmap for subject 1, trial 1.

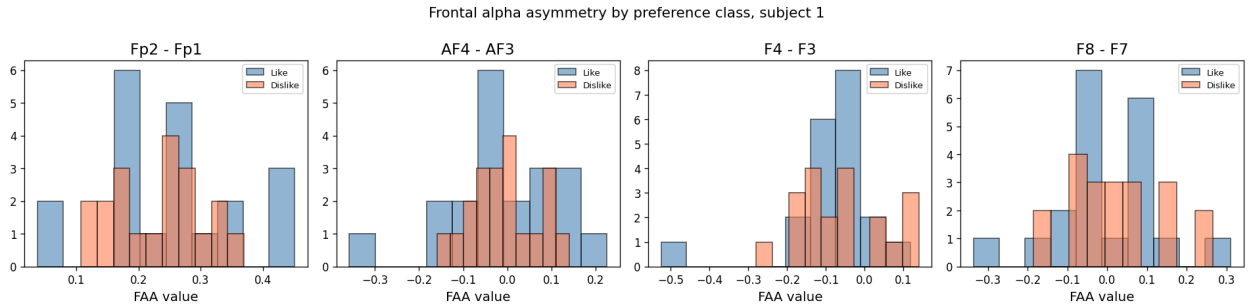


Figure 5: Distribution of FAA values for each frontal pair, split by binary preference label. Positive FAA indicates greater left frontal activation.

# 3 Classification

## 3.1 Methods

All classifiers are evaluated under 5-fold stratified cross-validation within each subject. Three strategies are compared:

1. Baseline: 164-dimensional features with median-split binarization. Three classifiers are trained independently (SVM with RBF kernel, XGBoost, and random forest), and the best per-subject result is reported.
2. Baseline with margin binarization: we do the same as above, but trials within ±0.5 rating units of the per-subject median are removed.
3. Majority-vote ensemble: XGBoost, random forest, and LightGBM combined by majority vote, after variance filtering and mutual-information feature selection retaining 50 features.

## 3.2 Results

Margin binarization improves mean accuracy from 62.4% to 64.5% (Table 2), confirming that removing ambiguous trials results in cleaner class boundaries. The majority-vote ensemble underperforms the single-model baseline (56.1%), likely because aggressive feature selection discards information. Inter-subject variability is substantial (range 35.3–83.3%), reflecting the highly individual nature of preference-related neural representations (Figure 6). Table 3 reports individual results; several subjects fall at or below chance, indicating that the feature set is not a catch-all for all individuals.

Table 2: Classification accuracy across 32 subjects (mean, standard deviation, and range).

| Strategy | Mean (%) | Std (%) | Range (%) |
|---|---|---|---|
| Baseline (median split) | 62.4 | 10.8 | 40.0 – 80.0 |
| Baseline with margin | 64.5 | 11.8 | 35.3 – 83.3 |
| Majority-vote ensemble | 56.1 | 11.1 | 30.0 – 77.5 |
| Chance | 50.0 | — | — |

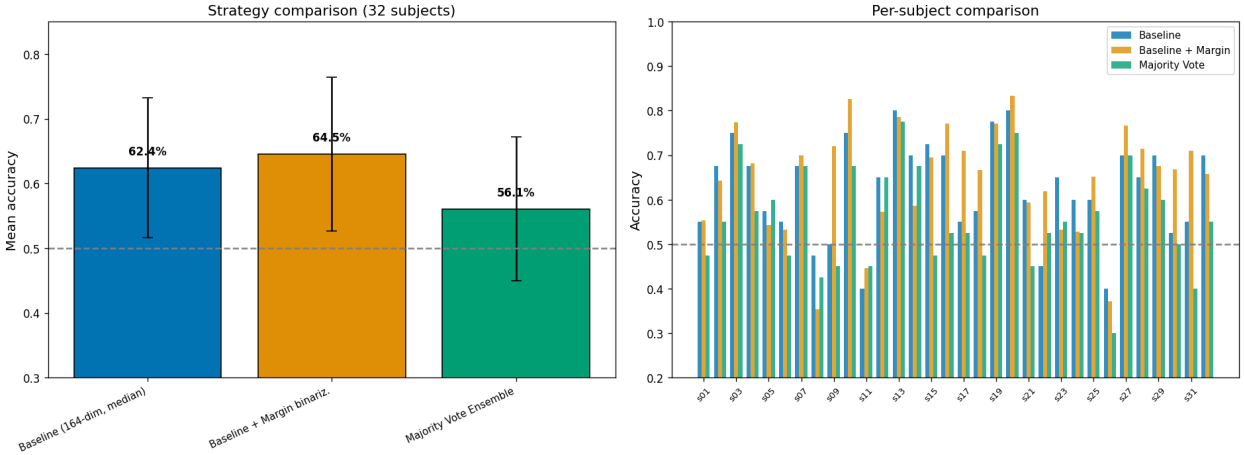


Figure 6: Per-subject accuracy for the three classification strategies.

Table 3: Per-subject classification accuracy (%) across three strategies. green ≥60%; yellow 50–59%; red <50%.

| Subj. | Baseline | Margin | Ensemble | Subj. | Baseline | Margin | Ensemble |
|---|---|---|---|---|---|---|---|
| s01 | 55.0 | 55.3 | 47.5 | s17 | 55.0 | 71.0 | 52.5 |
| s02 | 67.5 | 64.3 | 55.0 | s18 | 57.5 | 66.7 | 47.5 |
| s03 | 75.0 | 77.3 | 72.5 | s19 | 77.5 | 77.1 | 72.5 |
| s04 | 67.5 | 68.1 | 57.5 | s20 | 80.0 | 83.3 | 75.0 |
| s05 | 57.5 | 54.3 | 60.0 | s21 | 60.0 | 59.3 | 45.0 |
| s06 | 55.0 | 53.3 | 47.5 | s22 | 45.0 | 61.9 | 52.5 |
| s07 | 67.5 | 70.0 | 67.5 | s23 | 65.0 | 53.2 | 55.0 |
| s08 | 47.5 | 35.3 | 42.5 | s24 | 60.0 | 52.9 | 52.5 |
| s09 | 50.0 | 72.0 | 45.0 | s25 | 60.0 | 65.2 | 57.5 |
| s10 | 75.0 | 82.7 | 67.5 | s26 | 40.0 | 37.1 | 30.0 |
| s11 | 40.0 | 44.6 | 45.0 | s27 | 70.0 | 76.7 | 70.0 |
| s12 | 65.0 | 57.3 | 65.0 | s28 | 65.0 | 71.4 | 62.5 |
| s13 | 80.0 | 78.6 | 77.5 | s29 | 70.0 | 67.5 | 60.0 |
| s14 | 70.0 | 58.6 | 67.5 | s30 | 52.5 | 66.8 | 50.0 |
| s15 | 72.5 | 69.5 | 47.5 | s31 | 55.0 | 71.0 | 40.0 |
| s16 | 70.0 | 77.1 | 52.5 | s32 | 70.0 | 65.7 | 55.0 |

# 4 Interpretability & advanced analysis

## 4.1 SHAP feature importance

SHAP (SHapley Additive exPlanations) values were computed using a `TreeExplainer` (Lundberg and Lee 2017) on XGBoost models trained for each of the 32 subjects. Feature importance is estimated as the mean absolute SHAP value per feature, averaged across subjects.

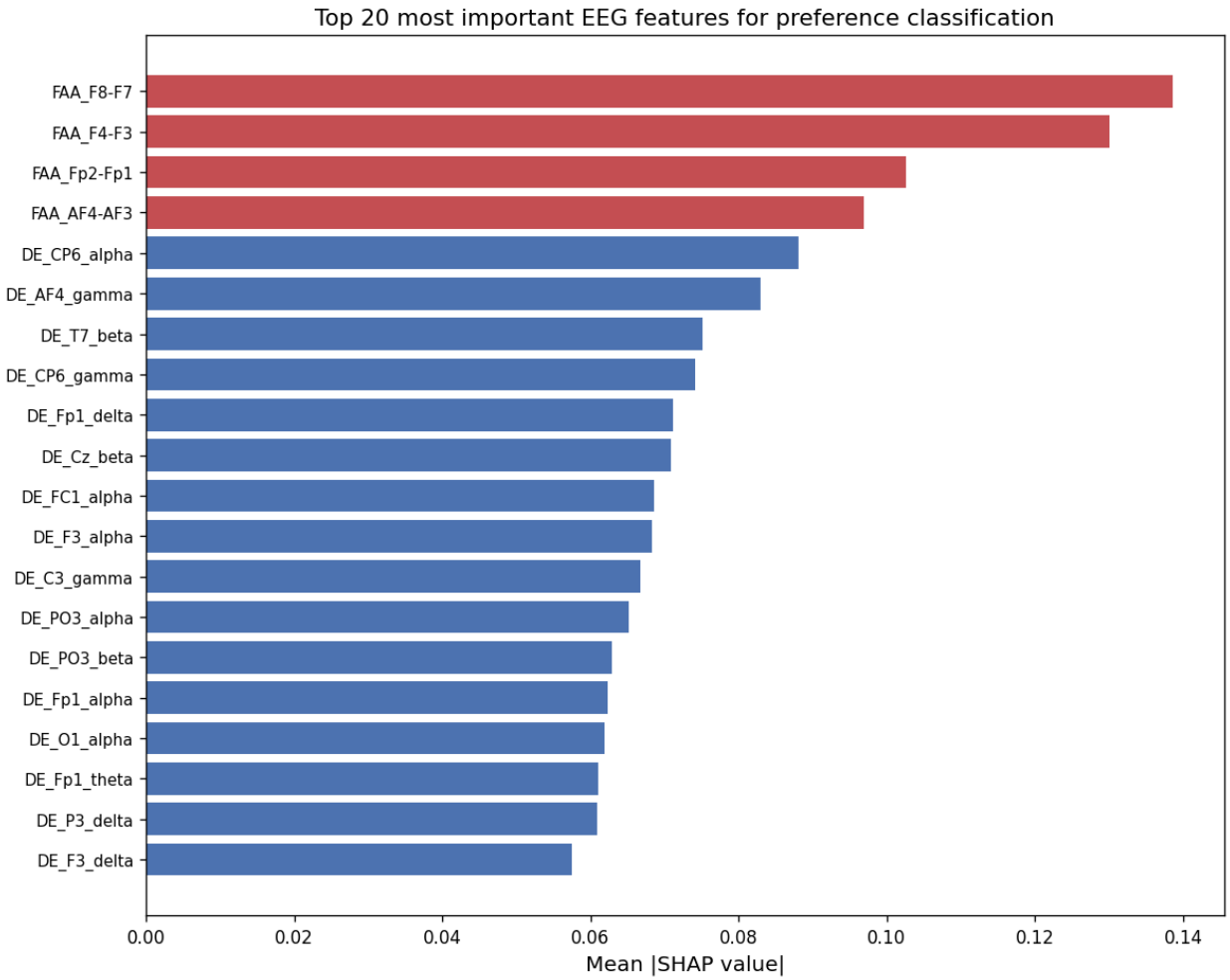


Figure 7: Top 20 features by mean absolute SHAP value, averaged across all 32 subjects.

Strikingly, all four FAA features occupy the top four positions in global importance (Table 4), with FAA at the F8–F7 pair ranking first. This constitutes strong support for the approach-withdrawal hypothesis, as frontal alpha asymmetry is not merely present but is the dominant class of predictor. The remaining top features are DE features spanning alpha, gamma, beta, and delta bands, concentrated at frontal, parietal, and temporal electrodes.

Table 4: Top 10 features by mean absolute SHAP value, averaged across all 32 subjects.

| Rank | Feature | Mean \|SHAP\| |
|---|---|---|
| 1 | FAA, F8–F7 | 0.138614 |
| 2 | FAA, F4–F3 | 0.130051 |
| 3 | FAA, Fp2–Fp1 | 0.102652 |
| 4 | FAA, AF4–AF3 | 0.096894 |
| 5 | DE, CP6, alpha | 0.088021 |
| 6 | DE, AF4, gamma | 0.082916 |
| 7 | DE, T7, beta | 0.075091 |
| 8 | DE, CP6, gamma | 0.074095 |
| 9 | DE, Fp1, delta | 0.071180 |
| 10 | DE, Cz, beta | 0.070823 |

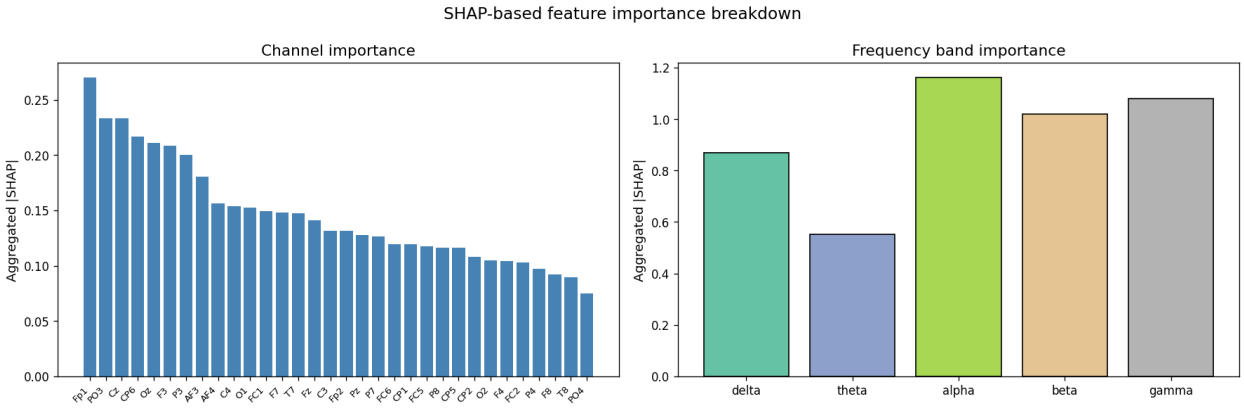


Figure 8: Aggregated SHAP importance by channel (left) and frequency band (right).

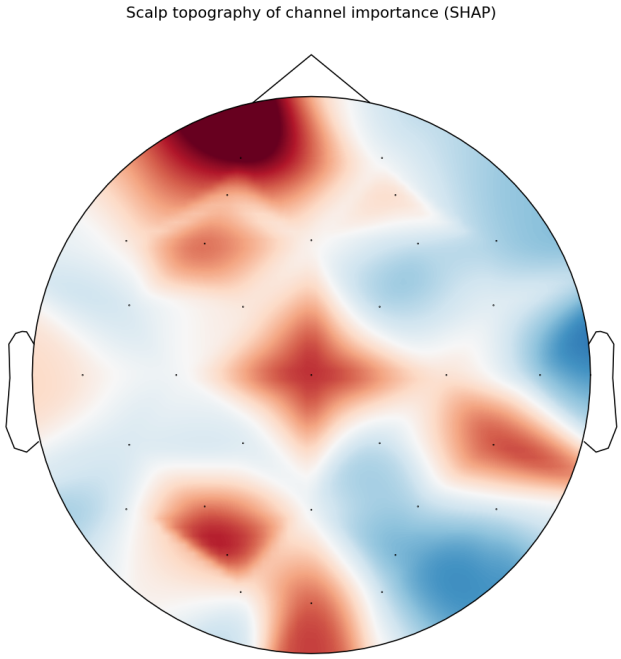


Figure 9: Scalp topographic map of channel-level SHAP importance summed across all frequency bands.

## 4.2 Time-resolved decoding

To identify when during a trial preference information becomes decodable, features were extracted from 5-second sliding windows with a 2.5-second step across the full 60-second trial for five representative subjects (s01, s03, s05, s08, s16).

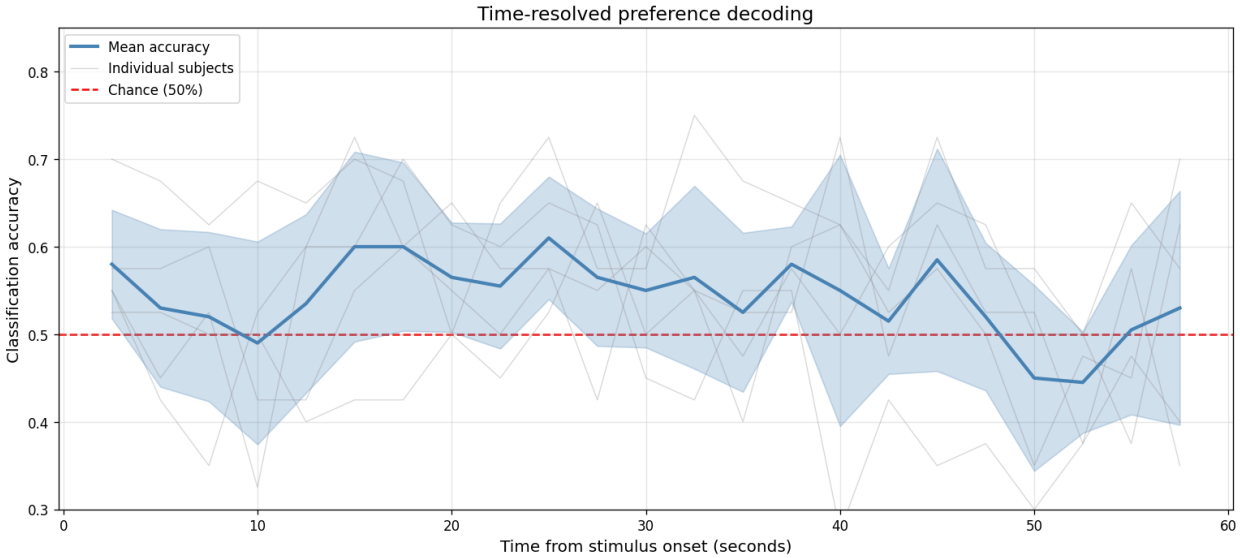


Figure 10: Time-resolved decoding accuracy for five representative subjects. Shading shows ±1 standard error across folds.

Decoding first exceeds the 55% threshold at approximately 2.5s after trial onset. Peak group accuracy is 61.0% at 25.0 seconds, after which accuracy declines, stabilizing at 53.0% at 57.5 seconds. The early onset suggests evaluation is rapidly effective, while the mid-trial peak and subsequent decline may reflect loss of or shifting attention over the course of the video clip.

## 4.3 Familiarity disentanglement

Song familiarity is a reasonable confound, since prior exposure to a track may strengthen preference-related neural signals independently of the current listening experience. Trials were split by familiarity rating into high- and low-familiarity subsets, and classification was repeated for all 32 subjects.

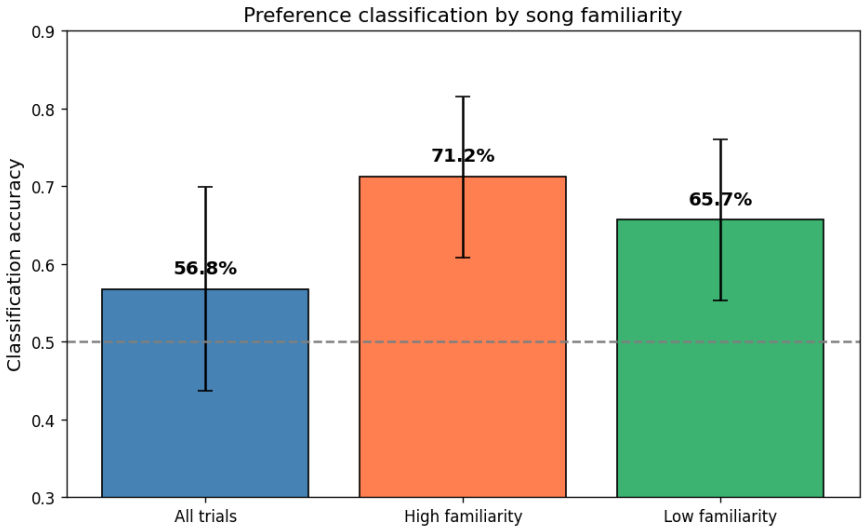


Figure 11: Classification accuracy under three familiarity conditions.

Table 5: Familiarity disentanglement results.

| Condition | Mean accuracy (%) | Std (%) | $n$ subjects |
|---|---|---|---|
| All trials | 56.8 | 13.2 | 32 |
| High familiarity only | 71.2 | 10.3 | 28 |
| Low familiarity only | 65.7 | 10.4 | 29 |

Both familiarity conditions substantially exceed the all-trials baseline (56.8%), suggesting that when the analysis is restricted to trials with clear familiarity status, classification improves markedly. The high-familiarity advantage over low-familiarity (71.2% versus 65.7%) is consistent with the hypothesis that prior exposure consolidates preference-related neural representations.

# 5 Discussion & summary

Binary music preference is decodable from EEG at above-chance accuracy when we are principled and intentional with our feature engineering. The best mean accuracy of 64.5% is achieved with margin binarization, compared to a 50% chance baseline, with individual subjects ranging from 35.3% to 83.3%.

The most striking finding is that all four FAA features occupy the top four positions in global SHAP importance, which constitutes strong support for the approach-withdrawal hypothesis. Frontal alpha asymmetry is not merely present but is the dominant class of predictor.

Preference information is decodable within the first 2.5s and peaks near 25.0 seconds (61.0%), after which accuracy declines toward 53.0% at trial end. The familiarity analysis shows that both high- and low-familiarity conditions improve over the all-trials baseline (71.2% and 65.7% versus 56.8%).

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